

Rahul Ashok Jeyalakshmi

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EDUCATION

State University of New York, United States

Sep 2023 – Jan 2025

Master of Science in Industrial Engineering

KPR Institute of Engineering and Technology, India

Jul 2018 – May 2022

Bachelor of Engineering in Mechanical Engineering

SKILLS

Mechanical Design & Analysis: SolidWorks, AutoCAD Mechanical, Fusion 360, ANSYS, Engineering Drawing, GD&T.

Industrial Skill: Lean Manufacturing, Six Sigma, 5S, Kaizen, Kanban, Process Optimization, Root Cause Analysis, Statistical Process Control.

Tools: MS Excel, Minitab.

PROFESSIONAL EXPERIENCES

MABS Associates

Jun 2022 – Mar 2023

Industrial Service Engineer – Trichy, India

- Diagnosed mechanical failures in industrial pumps and rotating equipment, including seal leakage, misalignment, and bearing wear.
- Applied root cause analysis (RCA) to improve equipment reliability, reducing downtime by 25%.
- Collaborated with team members to standardize troubleshooting procedures and improve service performance.
- Optimized maintenance processes, reducing repair time by 30% and achieving cost savings of up to ₹50,000.

Harihar Alloys Private Limited

Aug 2021 – Sep 2021

Production Engineer Intern- Viralimalai, India

- Monitored production processes and inspected forged components, reducing defects and production waste by 13%.
- Improved manufacturing efficiency by 15% through process optimization and workflow improvements.
- Supported production planning and scheduling to ensure smooth operations and timely delivery.
- Collaborated with cross-functional teams to monitor production schedules and coordinate timely deliveries.

Shree Vari Fly Ash Bricks

Jun 2020 – Feb 2021

Supply Chain Coordinator Assistant- Thoothukudi, India.

- Supported logistics planning and inventory coordination, achieving a 10% reduction in transportation costs and minimizing stock shortages.
- Coordinated with fly ash suppliers to maintain a reliable supply chain, negotiate favorable terms, and ensure uninterrupted material availability.
- Improved inventory accuracy by 15% through regular stock audits and enhanced record-keeping practices.
- Enhanced supply chain visibility by developing procurement and inventory tracking reports using Excel and ERP tools.

ACADEMIC PROJECTS

Analyzing the Environmental and Social Impact of Sustainable Practices

Sep 2023 - Dec 2023

- Evaluated the engineering feasibility of sustainable supply chain technologies in logistics operations.
- Evaluated carbon footprint reduction formulas and logistics strategies, including CO2 emissions comparisons across air, road, and sea transport.
- Assessed sustainable packaging opportunities using recycle rate and packaging efficiency metrics to reduce material usage and waste.
- Analyzed emerging green supply chain technologies, resulting in a 20% projected reduction in carbon emissions for the proposed E-commerce strategies.

Optimization of Retail Store Layout Using Market Basket Analysis and A-Score Efficiency

Sep 2023 - Dec 2023

- Conducted a Facility Layout Optimization for Saideep Superstore, enhancing product visibility and operational efficiency.
- Applied Market Basket Analysis to uncover hidden product associations and improve customer flow.
- Utilized A-Score efficiency method to evaluate layout effectiveness, improving layout efficiency from 61.5% to 81.4%.
- Implemented layout changes by reallocating aisles and rack positions based on customer preferences, improving shopping and product accessibility.

Process Control & Capability Analysis for Performance Optimization

Jan 2024 - Apr 2024

- Applied statistical process control techniques using X-bar and R charts to evaluate process stability and variation.
- Implemented process improvements and quality control measures to enhance consistency and operational efficiency.
- Conducted process capability analysis to assess performance against specification limits.
- Evaluated mean, standard deviation, and defect rates (PPM) to identify variability and improvement areas.